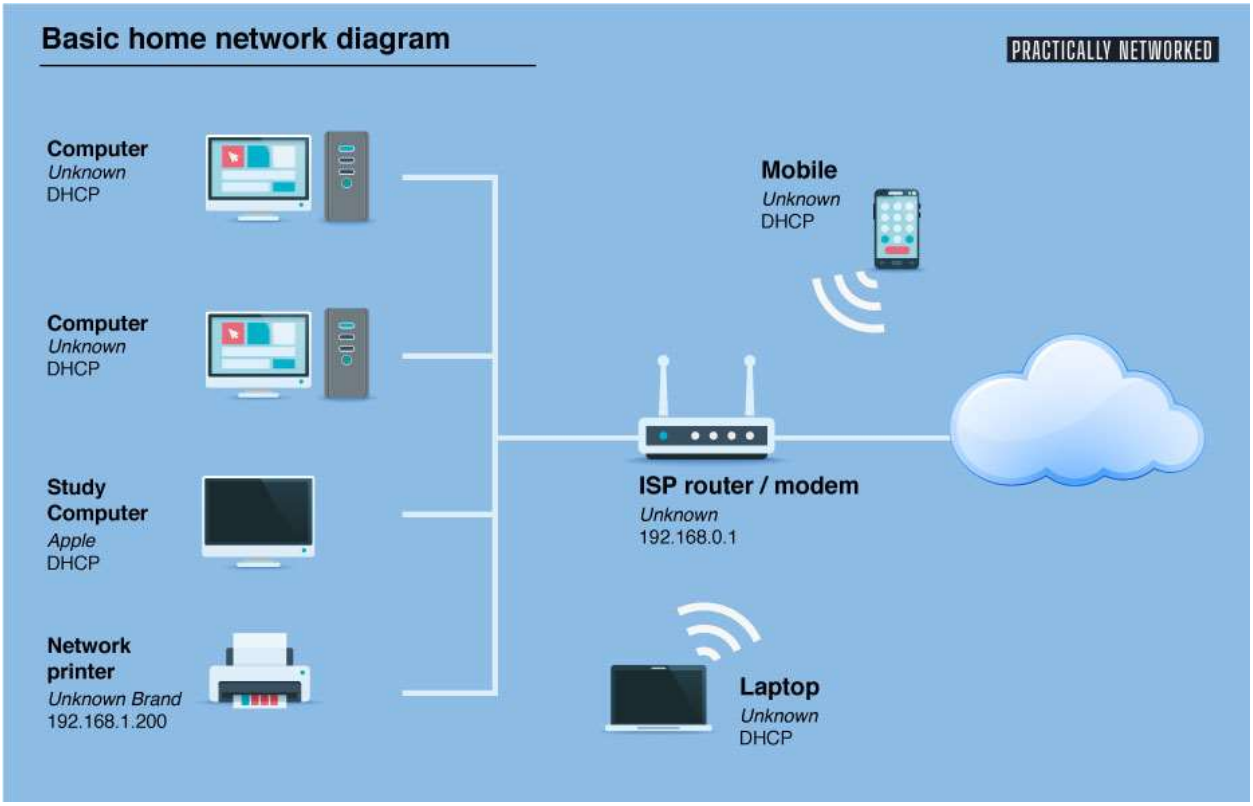
NETWORK DEVICES

TASK-1 Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.



NETWORK DEVICES

TASK-2 Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily (e.g., WiFi for streaming on YouTube, LAN gaming, etc.).

Network Device	Main Function	Real-World Use Case
Router	Connects different networks and provides internet access.	Connecting a home Wi-Fi network to the internet to watch YouTube videos.
Switch	Connects multiple devices within a LAN.	Connecting college computers for LAN gaming and file sharing.
Hub	Connects multiple devices and broadcasts received data to all ports.	Connecting older computers in a small lab for basic network communication.
Access Point (AP)	Provides wireless network access to devices.	Allowing smartphones and laptops to connect to Wi-Fi for Instagram, WhatsApp, and online classes.

NETWORK DEVICES

TASK-3 Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, access point).

1. Access Point: My phone connects to the home Wi-Fi through the access point, which sends my data into the local network.
2. Switch: The switch forwards data between connected devices within the local network and helps route traffic toward the router.
3. Router: The router forwards my order request from my home network to the internet.
4. Internet: The request reaches Zomato's servers, which process my order and send a confirmation back to my phone through the same network path.

NETWORK DEVICES

TASK-4 Given this situation: you have a Wi Fi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup Hint: Think about devices like switches, powerline adapters, or Wi Fi extenders.

1. Powerline Adapter

Uses the home's electrical wiring to carry network data from the router to the gaming PC. Connect one adapter to the router using Ethernet and plug it into a power socket. Connect the second adapter near the PC and use an Ethernet cable to connect the PC.

2. Wi-Fi Extender with Ethernet Port

Receives the router's Wi-Fi signal and extends it to a distant location. Connect the gaming PC to the extender's Ethernet port using an Ethernet cable.